Towards Understanding 3D Vision: The Role of Gaussian Curvature

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Introduction

What is this work about?

Introduction

Research Field

- 3D Scene Reconstruction
- 3D Scene Understanding

Approaches

- Stereo-Vision
  - **2021:** RAFT-Stereo
  - **2024:** Selective-IGEV
  - **2025:** FoundationStereo

Lipson, Lahav, Zachary Teed, and Jia Deng. "**RAFT-Stereo: Multilevel recurrent field transforms for stereo matching.**" 2021 International conference on 3D vision (3DV). IEEE, 2021.

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- Monocular Depth Estimation
  - **2024:** DepthAnythingV2
  - **2024:** DepthPro

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- ❌End-to-End Pipelines

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Introduction

Stereo-Vision

- Two views → Left + Right images
- Rectified Epipolar lines

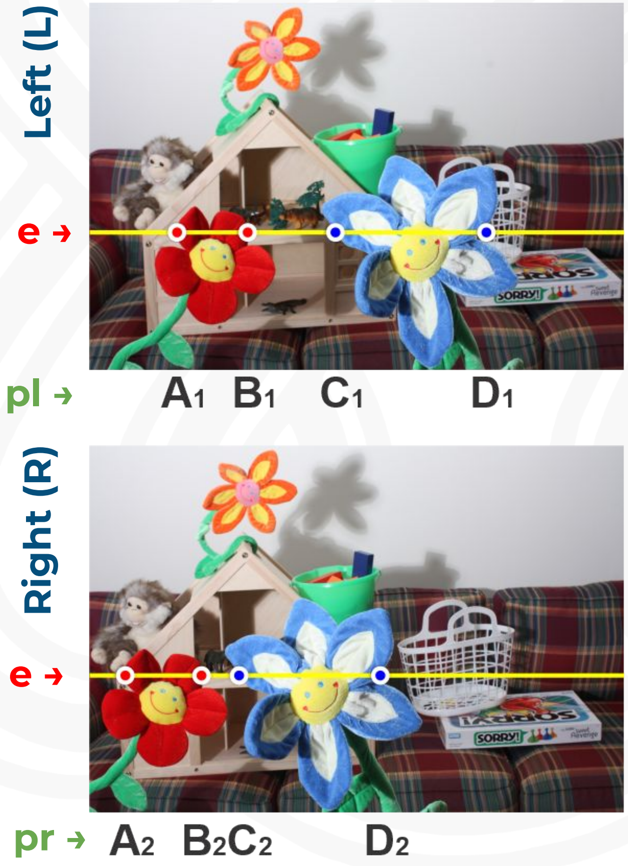
Left (L)

Right (R)

Introduction

Stereo-Vision

- Two views → Left + Right images
- Rectified Epipolar lines
- Feature ‘data’ Matching



Introduction

Stereo-Vision

- Two views → Left + Right images
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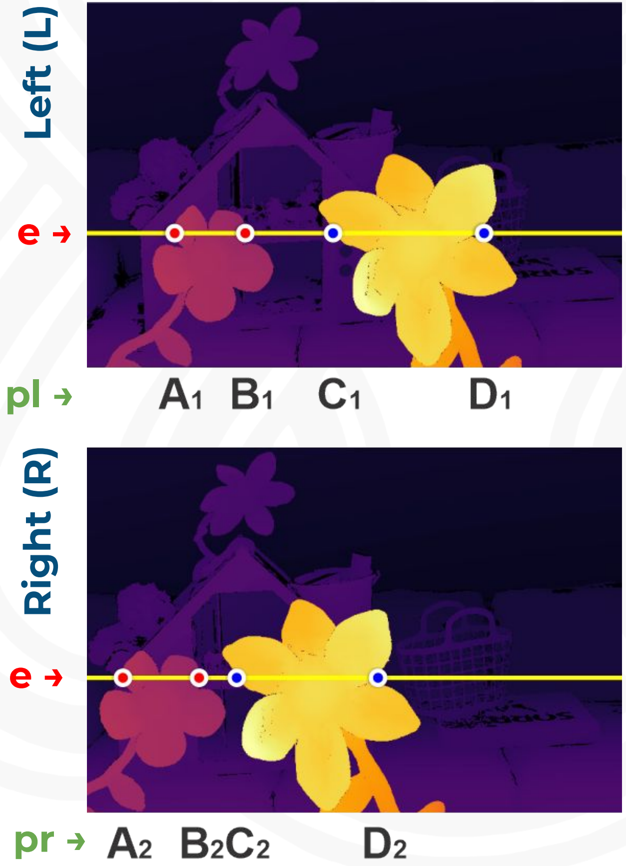
- Feature ‘data’ Matching
- Disparity Estimation

Disparity (d) = L(e, pl) - R(e, pr) >= 0

$$d(C) = L(e, C_1) - R(e, C_2)$$

$$d(A) = L(e, A_1) - R(e, A_2)$$

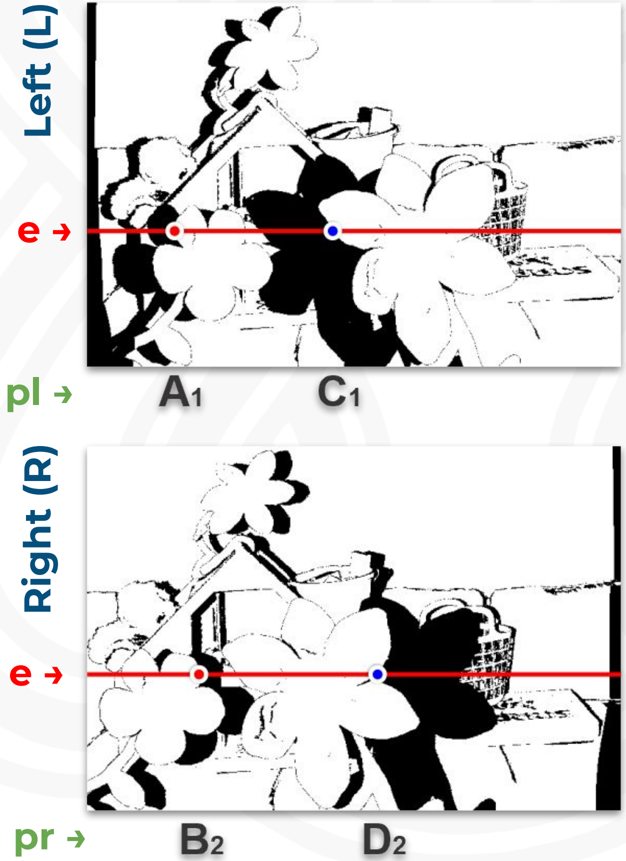
if d(C) > d(A) then →  → 



Introduction

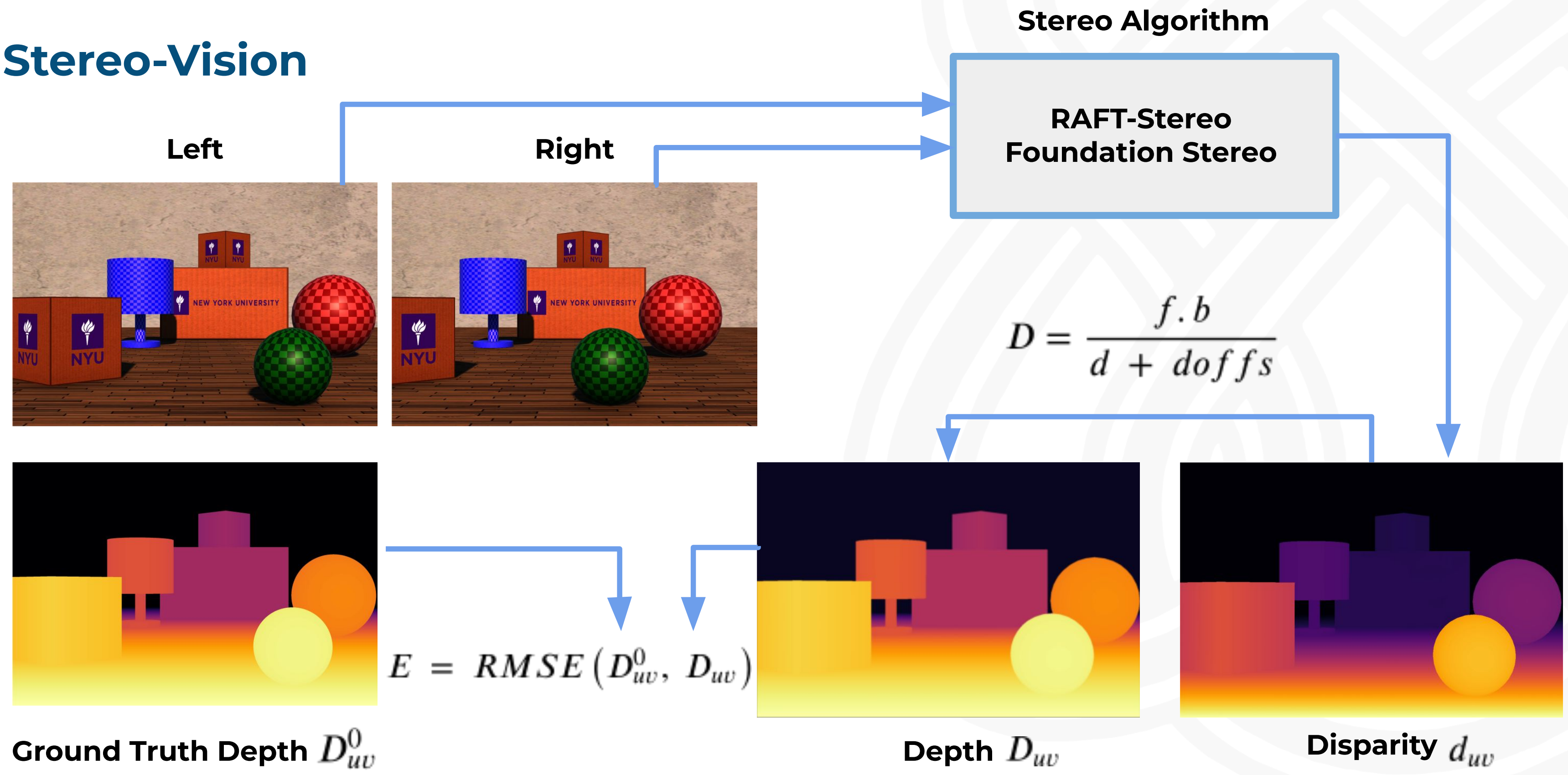
Stereo-Vision

- Two views → Left + Right images
- Rectified Epipolar lines
- Feature ‘data’ Matching
- Disparity Estimation
- Poor Data Matching
  - **Occlusions**
  - Textureless Regions
  - Repetitive Patterns



Introduction

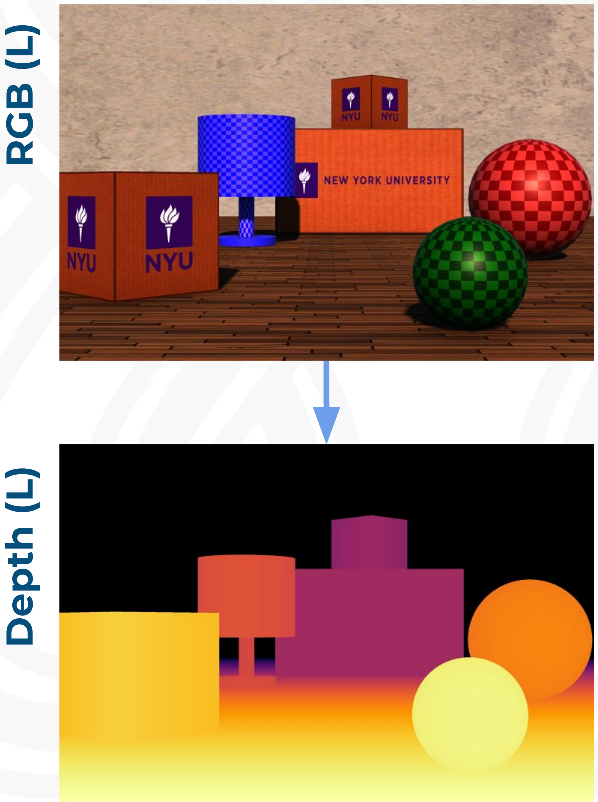
Stereo-Vision



Introduction

Monocular Depth Estimation (MDE)

- Single view → Depth
- Preserve scene structure
- ✅ Normals & Gradients
- ❌ Relative Depth



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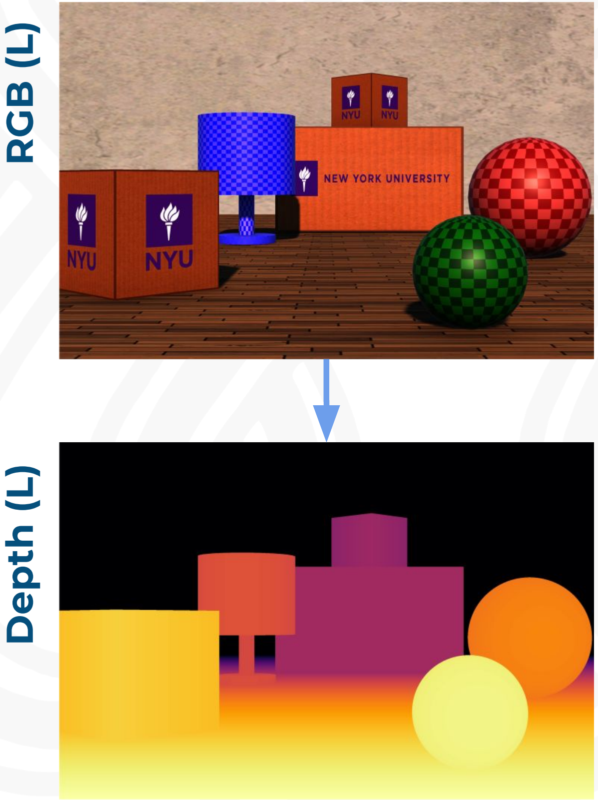


Research Questions

- What is the **distribution** of Gaussian curvature in real indoor 3D scenes?
- How do **SOTA methods** reconstruct depth, surface normals, and Gaussian curvature?
- What **role** does Gaussian curvature play in 3D reconstruction?

Gaussian Curvature

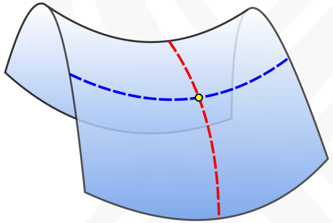
and its role in 3D Reconstruction

Gaussian Curvature

Gaussian Curvature (K)

- **Intrinsic property** of a surface;
- K=0 represents developable surfaces;
- The **sign** of K indicates the local shape of the surface;
- K is obtained by multiplying the **principal curvatures** k1 and k2;

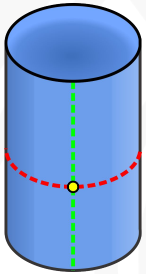
$$K = k_1 \cdot k_2$$



PC with Different signs

$(k1 > 0)$ and $(k2 < 0)$

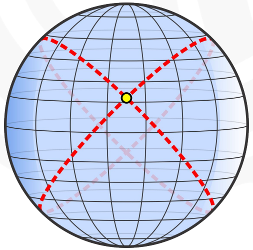
$K < 0$



One PC is Zero

$(k1 > 0)$ and $(k2 = 0)$

$K = 0$



PC with the same signs

$(k1 > 0)$ and $(k2 > 0)$

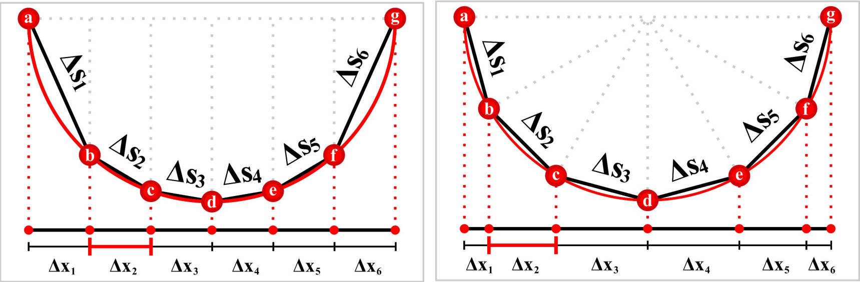
$K > 0$

Gaussian Curvature

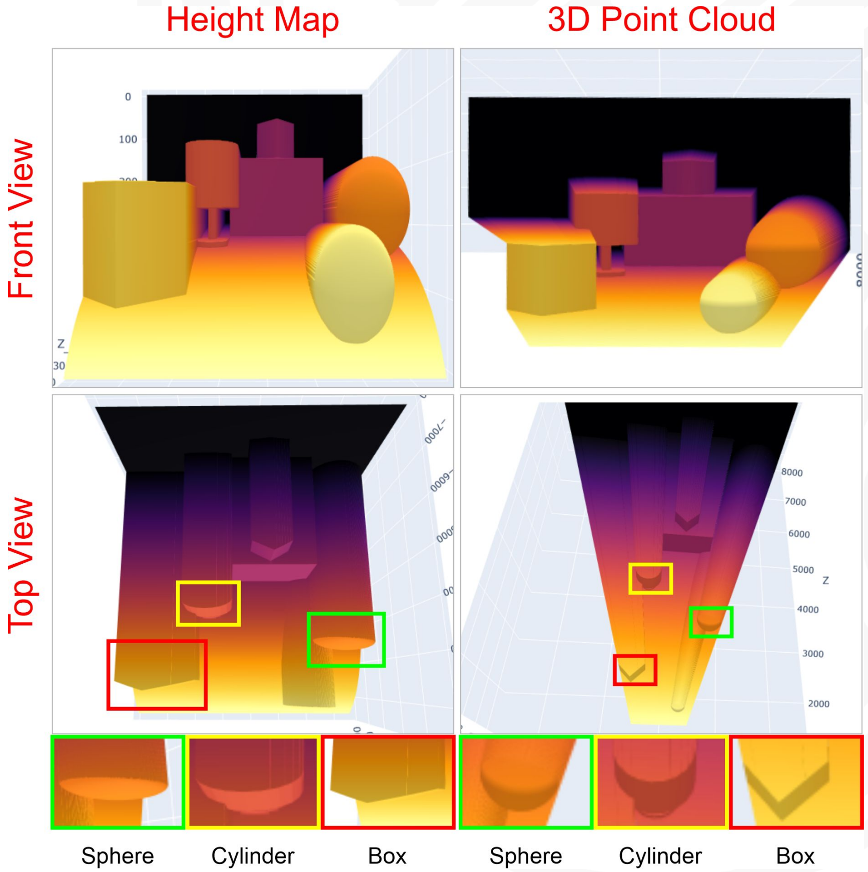
Gaussian Curvature (K)

- Height Map use a regular grid, then
  - it does not preserve surface geometry;
- We compute K on 3D Point Cloud;

Height Map **3D Point Cloud**



Δx is constant Δs is constant



Datasets

and Benchmarks

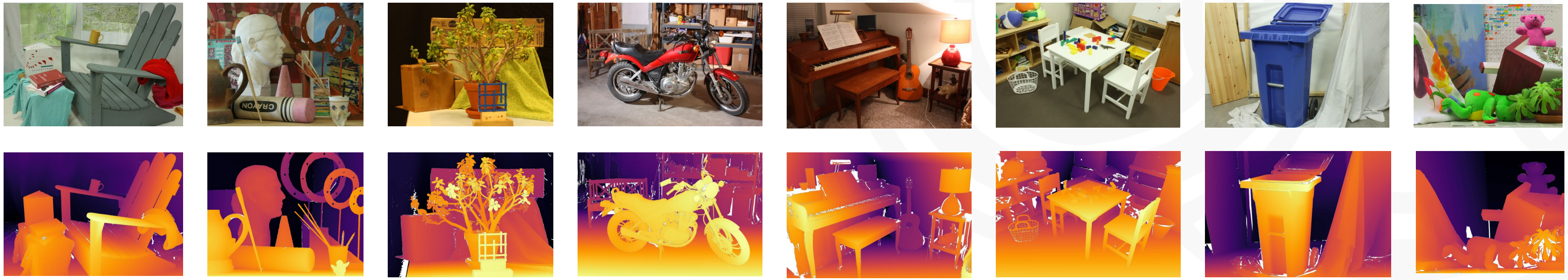
Datasets

Middlebury Dataset

- High resolution Ground Truth;
- 15 Training images;
  - Provides Techniques' results;
- 15 Test images;

	# of training images	# of test images	**Full resolution** up to 3000 x 2000 ndisp <= 800
Input data im0.png, im1.png, calib.txt	15	15	MiddEval3-data-F.zip (341 MB)
Ground truth for left view disp0GT.pfm, mask0nocc.png	15	—	MiddEval3-GT0-F.zip (194 MB)
***Ground truth for right view** disp1GT.pfm, mask1nocc.png	15	—	MiddEval3-GT1-F.zip (194 MB)
***Ground truth y-disparities** disp0GTy.pfm, disp1GTy.pfm	15	—	MiddEval3-GTy-F.zip (180 MB)

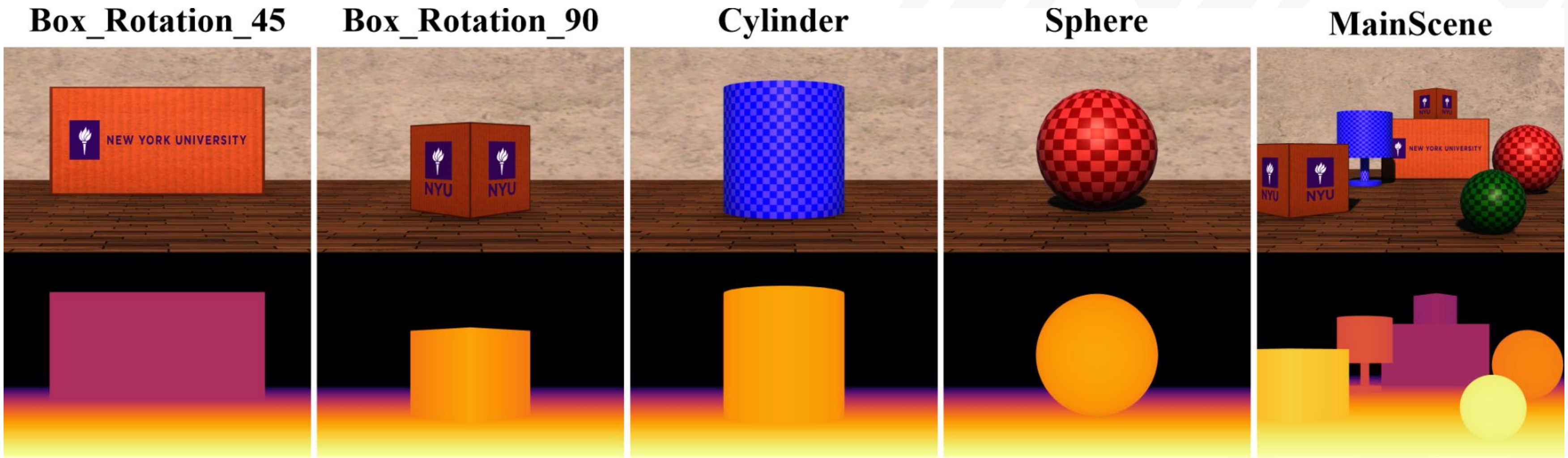
* these files are not needed for the evaluation



Datasets

3D Synthetic Scenes

- We created five synthetic scenes in a controlled environment to represent low K.
  - **Platform:** Unity 3D
  - **Sensor Size (H,W)** = (14.8, 22.2)mm
  - **Image Size (H,W)** = (2000, 3000)px
  - **Focal Length (mm, px)** = (35, 4729.73)
  - **Pixel Size** = 0.0074mm
  - **Baseline** = 200mm

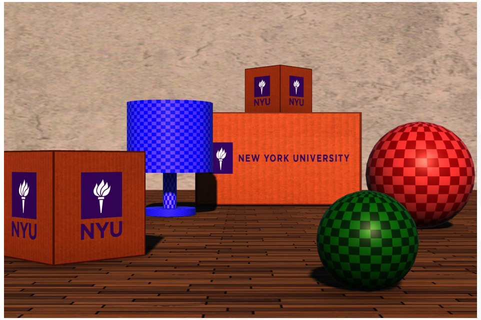


Datasets

3D Synthetic Scenes

- Sphere has $K = \frac{1}{r^2}$ everywhere;
- All the boxes, ground and wall have $K = 0$;
- Lamp is composed of cylinders, then it also has $K = 0$;

RGB (L)



Depth (L)

Spheres: Green r = 0.125m | Red r = 0.25m

Datasets

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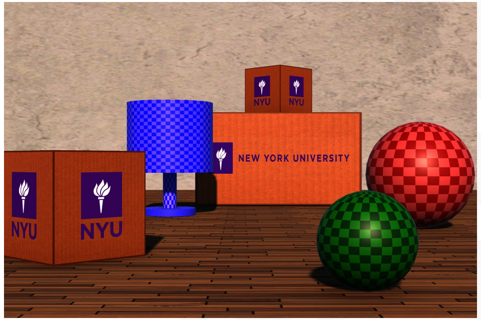

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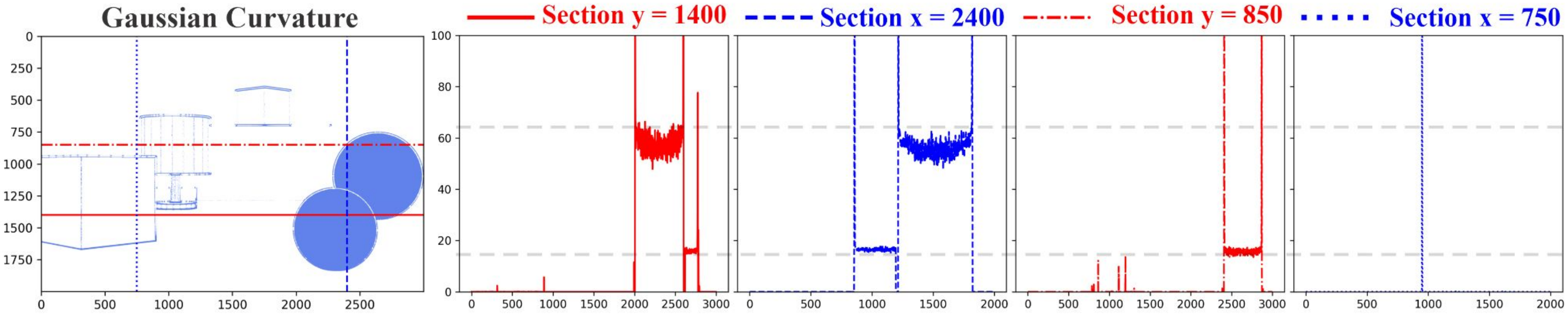


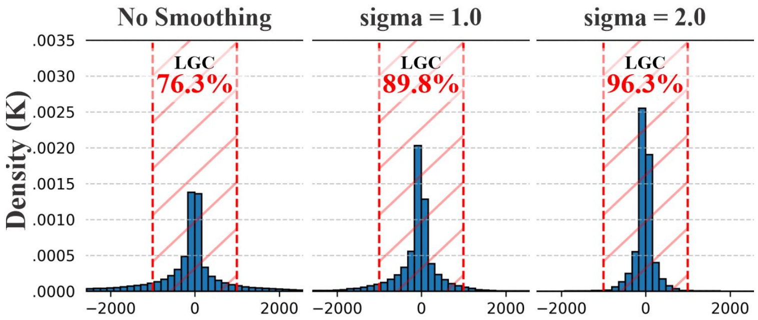
Results

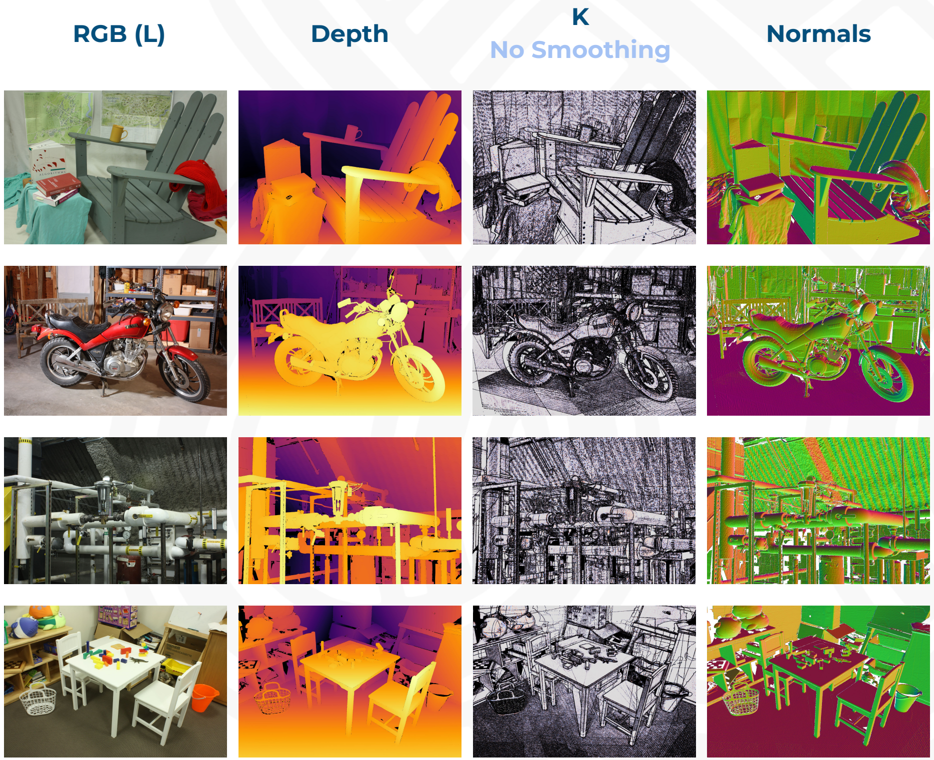
What do our results suggest?

Results (Part I)

Middlebury Dataset

- **Low Gaussian Curvature (LGC)**
- Percentage of **K**, where **-W > K < W**
- **LGC** over the 15 training images;
- White color represent low |K|;

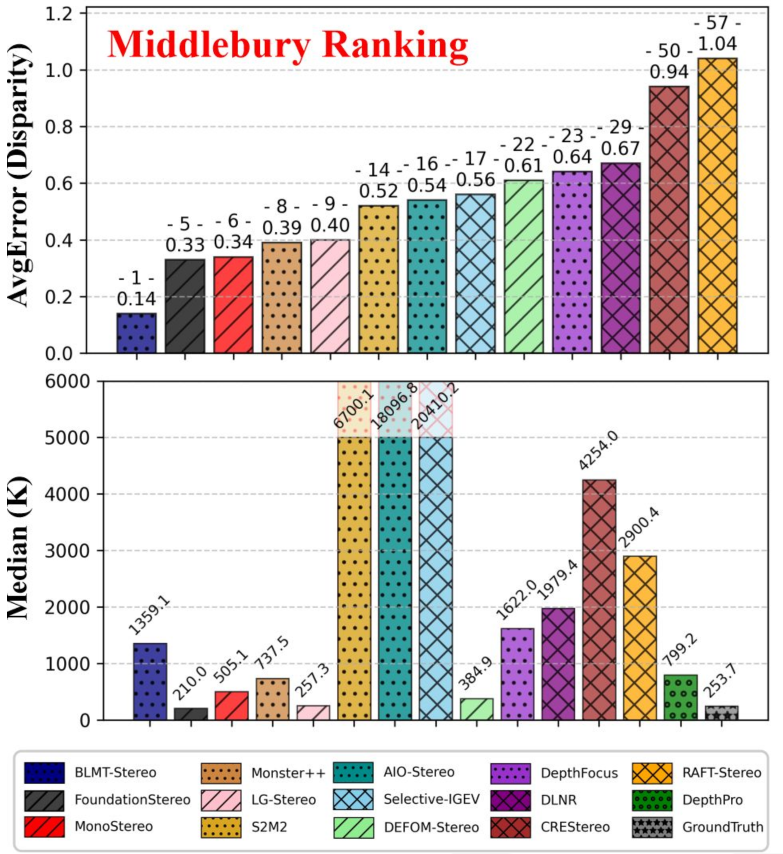




Results (Part I)

Middlebury Dataset

- **Average Disparity Error** over the 15 training images;
- Most of the best ranked approaches in the Middlebury Benchmark also predicts low $|K|$;
- Interestingly, most of them use **MDE features** in their pipelines;



Results (Part I)

Middlebury Dataset

- Results in the following table are sorted by **LGC** metric;
- **Superscripts** indicate each method's rank in the Benchmark;
- Darker cell shading highlights **better performance**: Top 1 , Top 3 , or Top 5 .

Technique	LGC	AvgError ↓	RMS ↓	Bad 2.0 ↓	Bad 4.0 ↓
FoundationStereo (Wen et al., 2025)	81.3	0.33^{2}	2.86^{4}	0.79^{2}	0.40^{3}
LG-Stereo	78.1	0.40^{5}	2.94^{5}	1.04^{5}	0.44^{4}
DEFOM-Stereo (Jiang et al., 2025)	69.2	0.61^{9}	3.66^{8}	2.26^{10}	1.19^{10}
MonoStereo (Cheng et al., 2025b)	64.6	0.34^{3}	2.46^{2}	0.94^{3}	0.35^{2}
MonSter++ (Cheng et al., 2025a)	56.5	0.39^{4}	2.54^{3}	1.17^{6}	0.47^{5}
BLMT-Stereo	43.1	0.14^{1}	1.33^{1}	0.17^{1}	0.09^{1}
DepthFocus	39.9	0.64^{10}	4.27^{11}	1.84^{7}	1.08^{9}
DLNR (Zhao et al., 2023)	36.3	0.67^{11}	3.90^{10}	2.92^{11}	1.41^{11}
RAFT-Stereo (Lipson et al., 2021)	29.6	1.04^{13}	5.25^{13}	5.25^{13}	2.89^{13}
CREStereo (Li et al., 2022)	25.8	0.94^{12}	5.21^{12}	4.01^{12}	2.04^{12}
S2M2 (Min et al., 2025)	17.5	0.52^{6}	3.73^{9}	1.00^{4}	0.59^{6}
AIO-Stereo (Zhou et al., 2025)	15.3	0.54^{7}	3.50^{6}	1.97^{8}	0.82^{7}
Selective-IGEV (Wang et al., 2024b)	14.0	0.56^{8}	3.57^{7}	2.18^{9}	0.92^{8}
Ground Truth	76.3	—	—	—	—

Results (Part I)

Middlebury Dataset

- **Normals Average Error** over the 15 training images (9 out of 15);
- Superscripts indicate the lowest 1-5th Normal Error (NormalsErr) per technique;
- Darker cell shading highlights **lower NormalsErr**: Top 1 , Top 3 , or Top 5 .

Technique	NormAvg ↓	Adirondack	ArtL	Jadeplant	Motorcycle	MotorcycleE	Piano	PianoL	Pipes
BLMT-Stereo	6.44	2.09[1]	18.08[4]	4.90[3]	6.31[1]	6.30[1]	2.93[1]	3.01[1]	8.36[1]
FoundationStereo	6.75	2.24[2]	15.92[1]	4.84[2]	7.88[4]	7.84[3]	3.70[3]	3.97[3]	9.04[3]
LG-Stereo	6.78	2.09[1]	17.97[3]	4.91[4]	7.00[2]	7.05[2]	3.36[2]	3.68[2]	8.46[2]
MonoStereo	7.29	2.32[3]	17.75[2]	4.60[1]	7.86[3]	8.32[4]	3.88[4]	4.63[4]	9.92[4]
MonSter++	8.13	2.70[4]	18.26[5]	5.05[5]	8.62	8.38[5]	4.81	5.86[5]	11.43
DEFOM-Stereo	8.18	2.77[5]	19.18	6.82	8.56[5]	8.70	4.71	6.18	11.31[5]
DepthFocus	8.93	4.24	19.38	10.49	8.79	8.83	4.61[5]	7.85	12.05
DLNR	9.05	3.50	19.76	7.20	9.40	9.50	6.22	8.88	11.60
S2M2	9.82	5.51	19.50	6.40	10.63	10.59	5.99	6.07	15.46
RAFT-Stereo	11.11	4.94	21.57	9.57	11.44	11.55	7.77	11.84	14.67
CREStereo	11.61	5.69	20.46	10.30	12.55	12.52	9.73	12.94	15.87
AIO-Stereo	16.65	10.03	22.05	12.28	15.49	15.55	14.60	16.45	23.45
Selective-IGEV	17.90	10.80	22.62	13.61	17.23	17.28	16.88	18.70	24.09
DepthPro	19.86	10.74	42.77	31.17	23.37	23.37	11.89	11.89	31.91

Results (Part I)

Middlebury Dataset

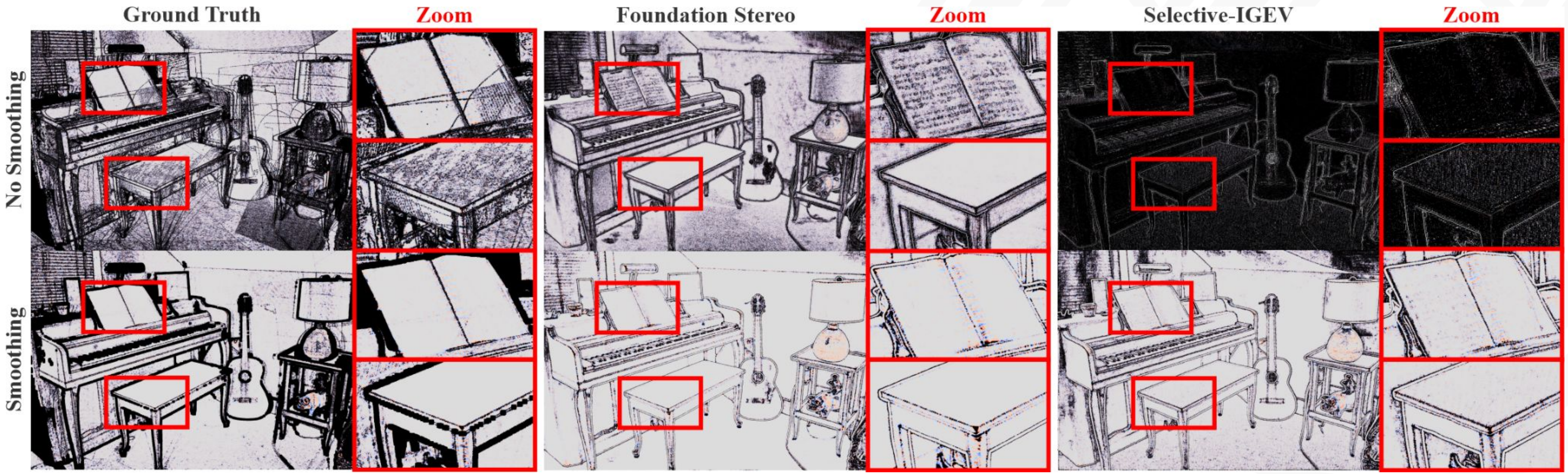
- **Normals Average Error** over the 15 training images (15 out of 15);
- Superscripts indicate the lowest 1-5th Normal Error (NormalsErr) per technique;
- Darker cell shading highlights **lower NormalsErr**: Top 1 , Top 3 , or Top 5 .

Technique	NormAvg ↓	Playroom	Playtable	PlaytableP	Recycle	Shelves	Teddy	Vintage
BLMT-Stereo	6.44	3.81^{1}	2.60^{1}	2.98^{1}	1.86^{1}	4.40^{1}	25.92^{2}	3.08^{1}
FoundationStereo	6.75	4.91^{3}	3.09^{3}	3.52^{4}	2.74^{5}	5.03^{3}	23.23^{1}	3.37^{3}
LG-Stereo	6.78	4.81^{2}	2.92^{2}	3.24^{2}	2.05^{2}	4.90^{2}	25.98^{3}	3.24^{2}
MonoStereo	7.29	5.24^{4}	3.14^{4}	3.36^{3}	2.36^{3}	6.33^{4}	26.02^{4}	3.64^{4}
MonSter++	8.13	6.04^{5}	4.05	4.00^{5}	2.76	9.57	26.26^{5}	4.19
DEFOM-Stereo	8.18	6.53	3.90^{5}	4.15	2.68^{4}	7.16^{5}	26.43	3.66^{5}
DepthFocus	8.93	6.97	4.01	4.17	3.43	7.35	27.34	4.39
DLNR	9.05	6.92	4.20	4.47	3.36	8.68	26.65	5.42
S2M2	9.82	8.02	5.82	5.94	4.58	8.54	28.00	6.20
RAFT-Stereo	11.11	8.45	5.90	6.16	4.38	12.06	27.08	9.33
CREStereo	11.61	9.78	7.80	7.84	5.19	10.17	26.95	6.43
AIO-Stereo	16.65	13.21	14.20	14.81	10.46	17.65	28.48	20.99
Selective-IGEV	17.90	14.31	15.87	16.44	11.57	18.15	28.98	22.00
DepthPro	19.86	18.42	8.87	9.17	9.62	14.88	42.35	7.46

Results (Part I)

Middlebury Dataset

- Despite good disparity/depth estimation, Selective-IGEV has **highest |K|** or **lower LGC**, and the **worst NormalsErr** among the techniques;



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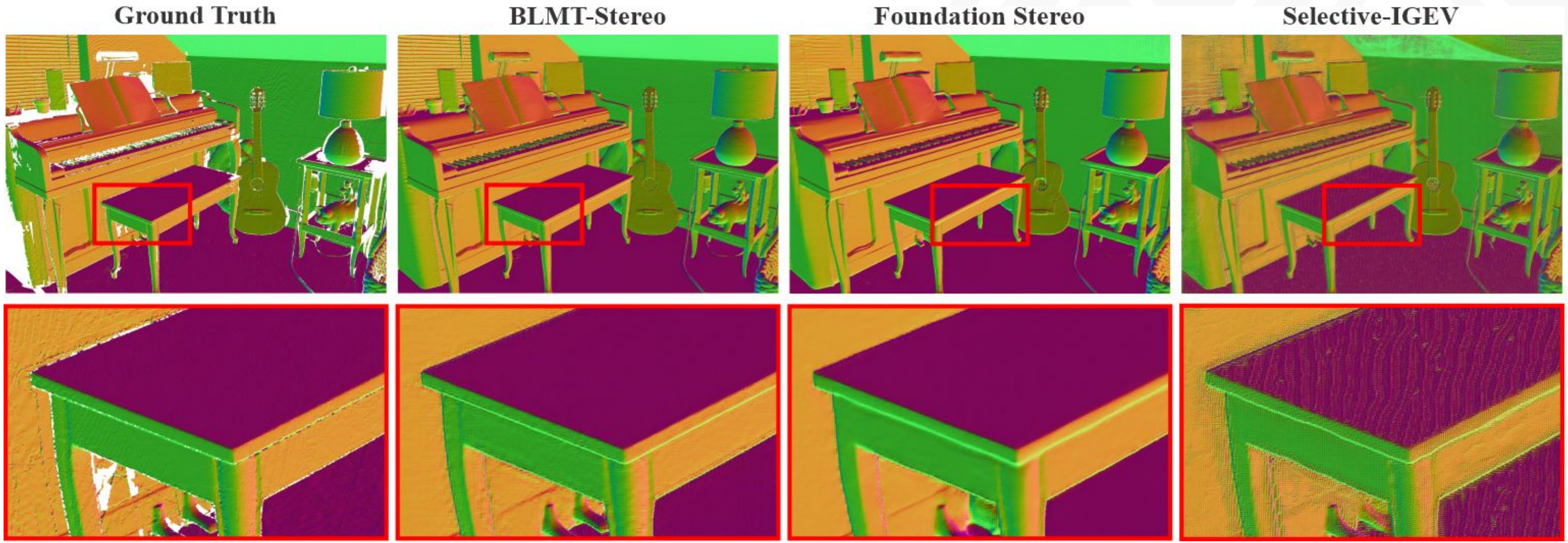

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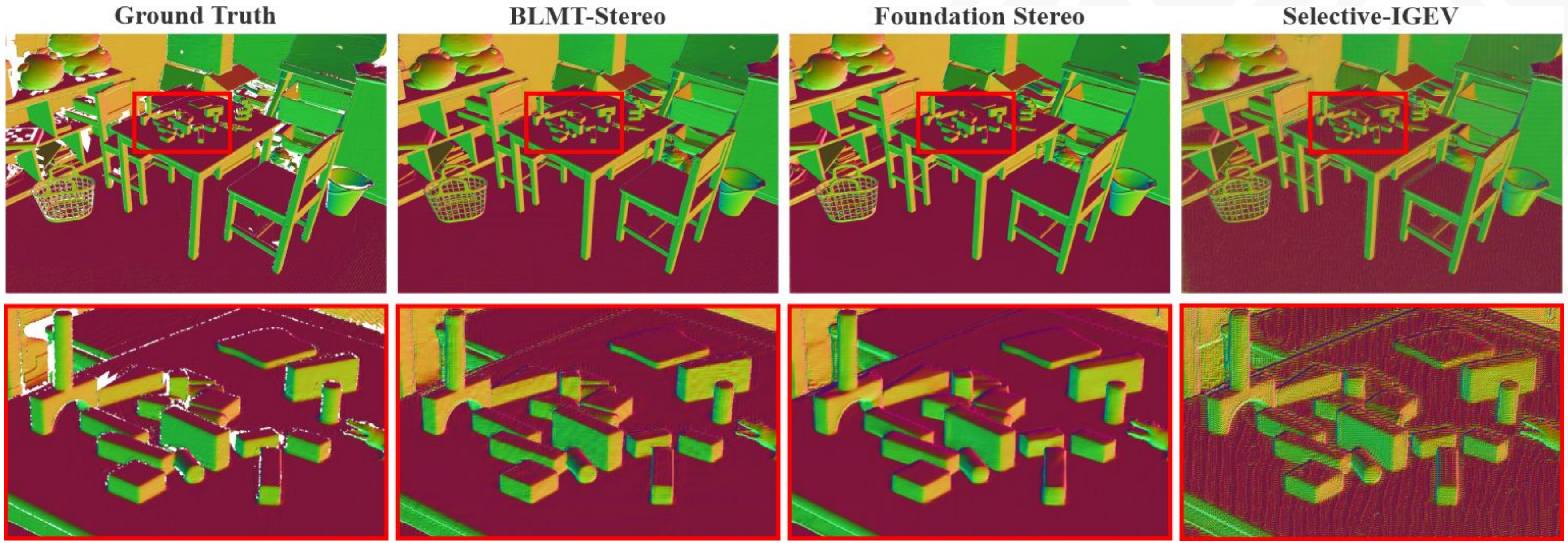
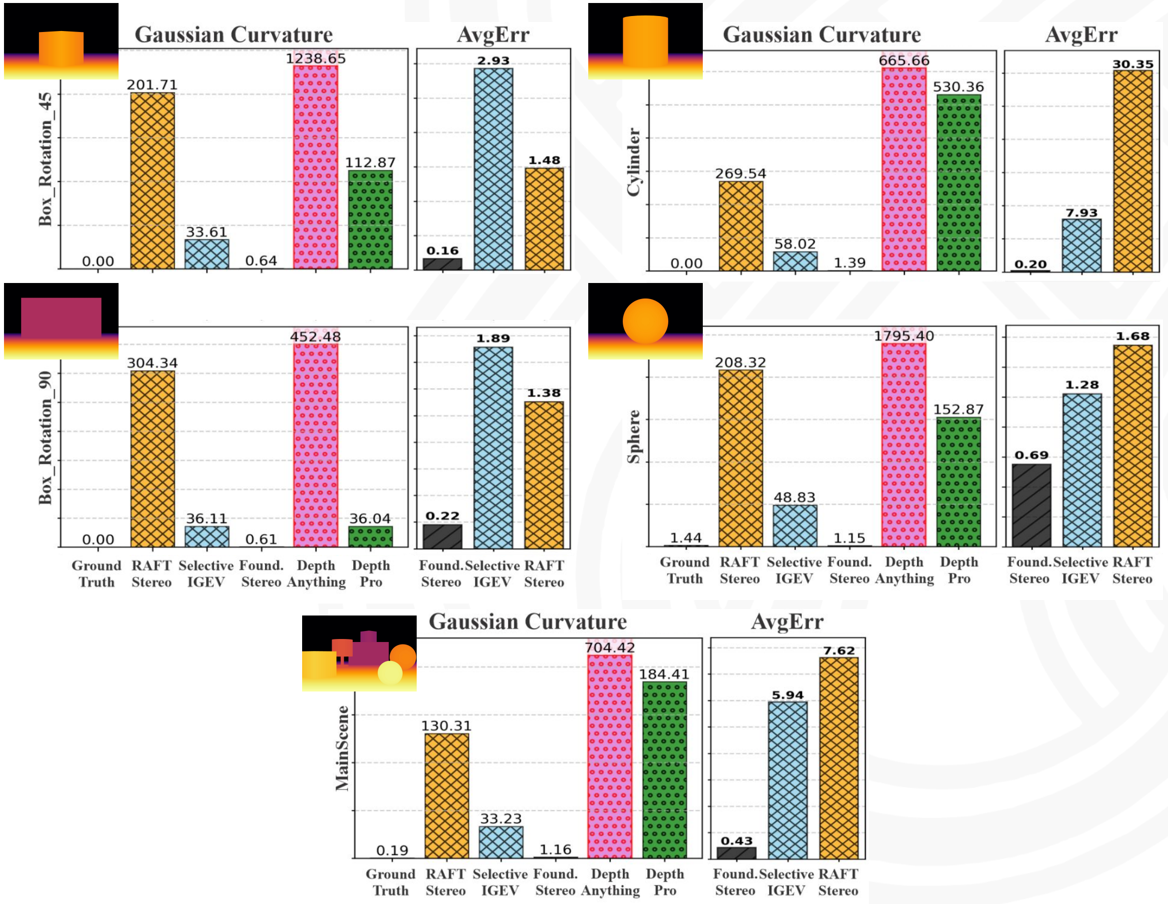


Results (Part II)

3D Synthetic Scenes

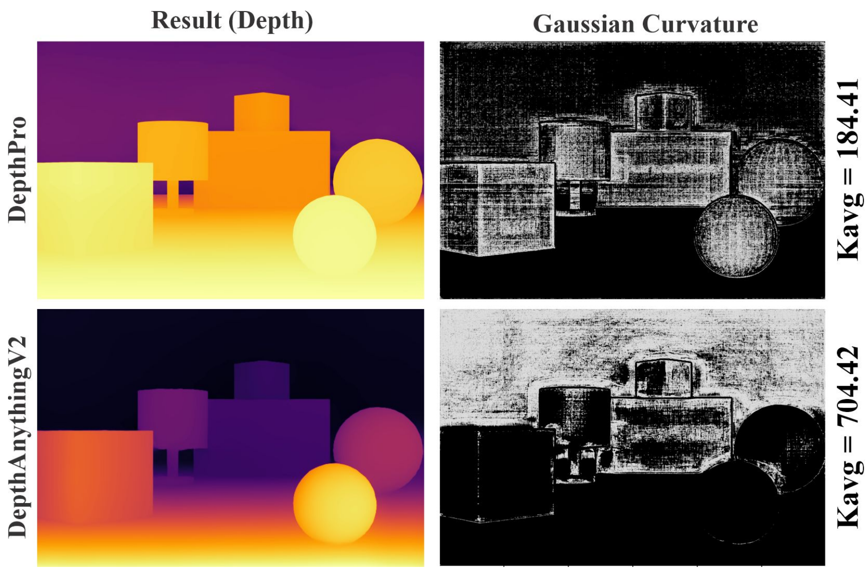
- **FoundationStereo** has the best depth AvgErr;
  - Error < 1 cm
- **FoundationStereo** has the lowest Avg |K|;
- **Selective-IGEV** and **RAFT-Stereo** have depth AvgErr varying from ~2 to ~30 cm;
- **DepthPro** outputs a 3D scene with Avg |K| lower than **DepthAnythingV2**.

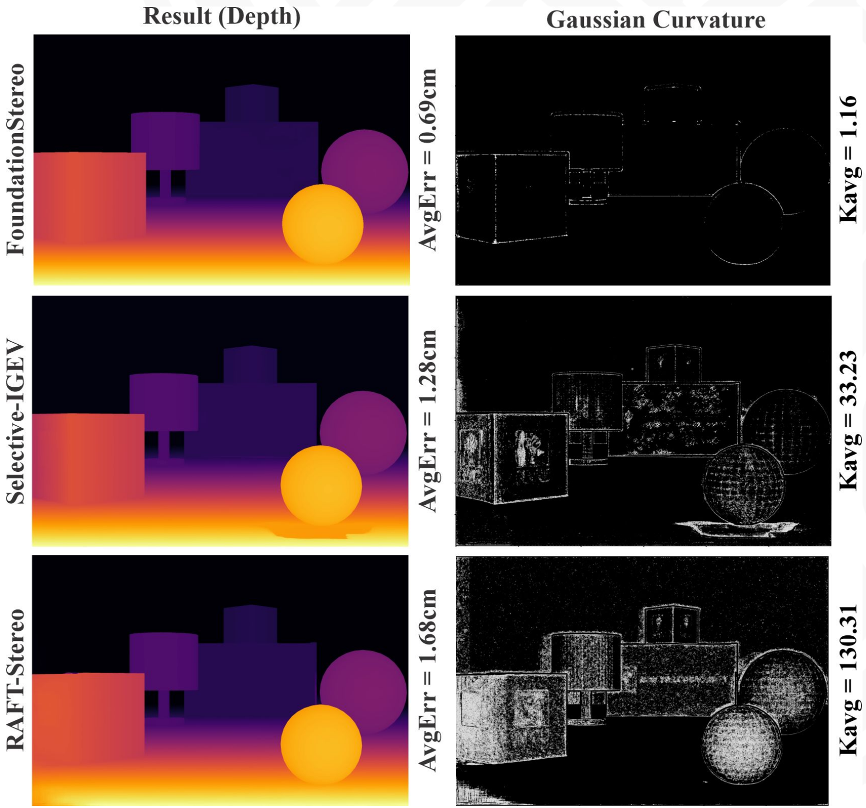


Results (Part II)

3D Synthetic Scenes

- Black color represent low |K|;
- FoundationStereo: Err = 0.69 & |K| = 1.16
- DepthPro: |K| = 184.41





Conclusion

Conclusion

Our Main Contributions

- We took a step towards 3D Scene Understanding
- We proposed a supervised metric, named as LGC

Answers

1) What is the distribution of Gaussian curvature in real indoor 3D scenes?
2) How do SOTA methods reconstruct depth, surface normals, and Gaussian curvature?
3) What role does Gaussian curvature play in 3D reconstruction?

Limitations

- Indoor scenes | 15 images

Future Directions

- Implement regularizer based on low |K|;
- Integrate it into a 3D Reconstruction pipeline;

Acknowledgements

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Thank you!

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Project Page

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